

BENCHMARK: NON-DETECTION

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2456230.8734 +/- 0.0046 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.008 +/- 0.03                  |
| Transit depth (Rp/Rs)^2                           | 0.0067 +/- 0.049 [%]            |
| Orbital Inclination (inc)                         | 86.14 +/- 1.45                  |
| Airmass coefficient 1 (a1)                        | 1.227 +/- 0.027                 |
| Airmass coefficient 2 (a2)                        | -0.1202 +/- 0.0025              |
| Scatter in the residuals of the lightcurve fit is | 1.97 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 3.44                            |
| Optimal Annulus                                   | 15.24                           |
| Transit Duration (day)                            | 0.1299 +/- 0.008                |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.008 ± 0.03 vs 0.1036 (deviation 2.33σ) |
| Duration fit vs published | 0.1299 d vs 0.1567 d (deviation 2.45σ)   |
| Mid-time O–C              | 11.3 min                                 |
| Depth SNR                 | 0.2                                      |
| Residual scatter          | 1.97 %                                   |

Light curve

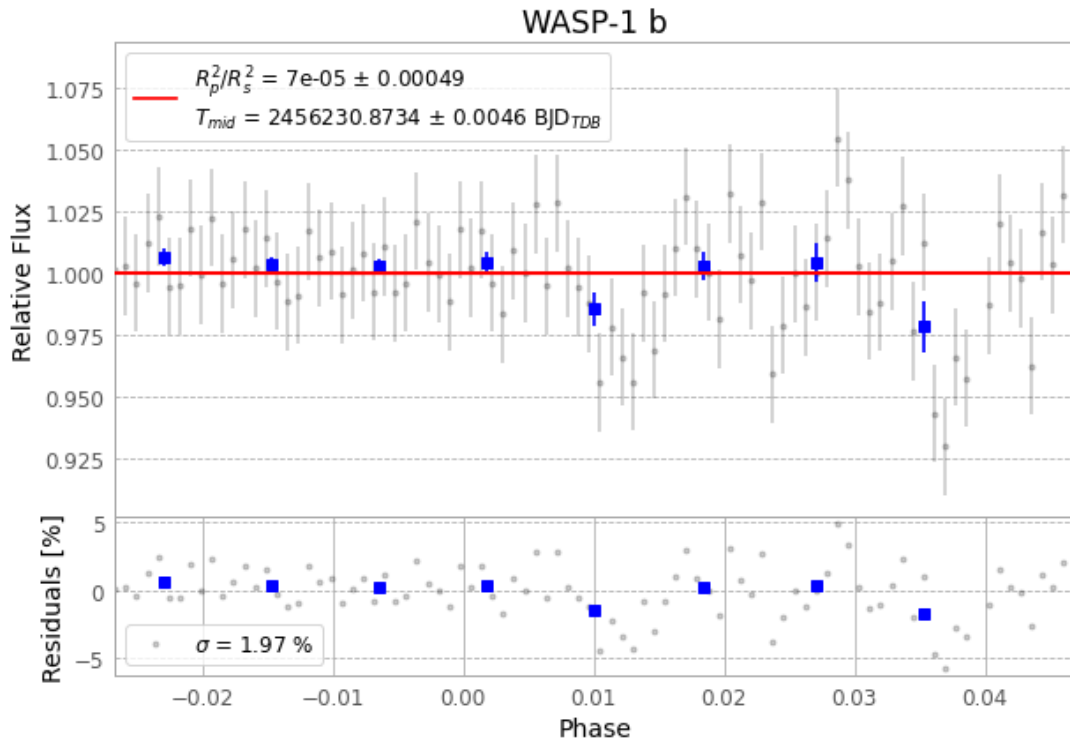


Figure 1 — FinalLightCurve\_WASP-1 b\_2012-10-30.png (SHA-256 283f7dc51d8382a9...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [244, 281], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 e2731fa3014295a0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

90 FITS frames (59 MB), WASP-1121030071212.FITS → WASP-1121030113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-121030.log (da54287cb61d...), FinalLightCurve\_WASP-1 b\_2012-10-30.png (283f7dc51d83...), FinalLightCurve\_WASP-1 b\_2012-10-30.csv (ffa59a09fd44...)

## Compute resources

Wall time 147.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:26Z.